

Question 13

```
vedsk0831@DESKTOP-83FU7HB:~$ cd ~/oss
nano q13.c
vedsk0831@DESKTOP-83FU7HB:~/oss$ echo "Hello from Q13 input file." > input.txt
vedsk0831@DESKTOP-83FU7HB:~/oss$ echo "This data will be redirected." >> input.txt
vedsk0831@DESKTOP-83FU7HB:~/oss$ cat input.txt
Hello from Q13 input file.
This data will be redirected.
vedsk0831@DESKTOP-83FU7HB:~/oss$ gcc -Wall -Wextra q13.c -o q13
vedsk0831@DESKTOP-83FU7HB:~/oss$ ./q13
Input redirected from input.txt
Output redirected to output.txt
vedsk0831@DESKTOP-83FU7HB:~/oss$ cat output.txt
Hello from Q13 input file.
This data will be redirected.
vedsk0831@DESKTOP-83FU7HB:~/oss$ ls -l input.txt output.txt
-rw-r--r-- 1 vedsk0831 vedsk0831 57 Sep 26 03:28 input.txt
-rw-r--r-- 1 vedsk0831 vedsk0831 57 Sep 26 03:28 output.txt
vedsk0831@DESKTOP-83FU7HB:~/oss$ |
```

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    int input_fd;
```

```
    int output_fd;
```

```
    pid_t pid;
```

```
    /* Open input file */
```

```
    input_fd = open("input.txt", O_RDONLY);
```

```
if (input_fd == -1)
{
    perror("Error opening input.txt");
    return 1;
}

/* Create output file */
output_fd = open("output.txt",
                O_WRONLY | O_CREAT | O_TRUNC,
                0644);

if (output_fd == -1)
{
    perror("Error opening output.txt");
    close(input_fd);
    return 1;
}

pid = fork();

if (pid == -1)
{
    perror("fork");
    return 1;
}

if (pid == 0)
```

```
{  
  
    /*  
     * Redirect stdin from input.txt  
     */  
    if (dup2(input_fd, STDIN_FILENO) == -1)  
    {  
        perror("dup2 stdin");  
        exit(1);  
    }  
  
    /*  
     * Redirect stdout to output.txt  
     */  
    if (dup2(output_fd, STDOUT_FILENO) == -1)  
    {  
        perror("dup2 stdout");  
        exit(1);  
    }  
  
    close(input_fd);  
    close(output_fd);  
  
    /*  
     * cat reads from stdin  
     * and writes to stdout  
     */  
    execlp("cat", "cat", NULL);
```

```
        perror("execlp");
        exit(1);
    }

    /* Parent */
    close(input_fd);
    close(output_fd);

    waitpid(pid, NULL, 0);

    printf("Input redirected from input.txt\n");
    printf("Output redirected to output.txt\n");

    return 0;
}
```